

Credit Risk Analysis

AUTHOR

Karabo Sishuba

```
library(tidyverse)

customer_data <- read_csv("C:/Users/Dell/Downloads/customer_cleaned_rows.csv")

anova_model <- aov(credit_limit ~ balance_tier, data = customer_data)
summary(anova_model)
```

```
              Df    Sum Sq   Mean Sq F value Pr(>F)
balance_tier    2 2.594e+10 1.297e+10   1254 <2e-16 ***
Residuals   8947 9.256e+10 1.035e+07
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
tukey_results <- TukeyHSD(anova_model)
print(tukey_results)
```

Tukey multiple comparisons of means
95% family-wise confidence level

```
Fit: aov(formula = credit_limit ~ balance_tier, data = customer_data)
```

```
$balance_tier
```

	diff	lwr	upr	p adj
Low Balance-High Balance	-6589.5066	-6898.0473	-6280.9659	0
Medium Balance-High Balance	-5829.4209	-6145.2470	-5513.5947	0
Medium Balance-Low Balance	760.0858	592.0523	928.1192	0

```
credit_summary <- customer_data %>%
  group_by(balance_tier) %>%
  summarise(
    count = n(),
    mean_credit = mean(credit_limit, na.rm = TRUE),
    sd_credit = sd(credit_limit, na.rm = TRUE),
    median_credit = median(credit_limit, na.rm = TRUE)
  )
print(credit_summary)
```

```
# A tibble: 3 × 5
  balance_tier count mean_credit sd_credit median_credit
  <chr>      <int>      <dbl>    <dbl>         <dbl>
1 High Balance    682    10287.    3494.         9500
2 Low Balance   4800     3697.    3234.         2500
3 Medium Balance 3468     4457.    3135.         3500
```

```
t_test_subset <- customer_data %>%
  filter(balance_tier %in% c("Low Balance", "High Balance"))

t_test_result <- t.test(purchases ~ balance_tier, data = t_test_subset, var.equal = FALSE)
print(t_test_result)
```

Welch Two Sample t-test

data: purchases by balance_tier

t = 6.6119, df = 699.17, p-value = 7.533e-11

alternative hypothesis: true difference in means between group High Balance and group Low Balance is not equal to 0

95 percent confidence interval:

739.3111 1363.8178

sample estimates:

	mean in group High Balance	mean in group Low Balance
	1897.2150	845.6506

```
purchases_summary <- t_test_subset %>%
  group_by(balance_tier) %>%
  summarise(
    count = n(),
    mean_purchases = mean(purchases, na.rm = TRUE),
    sd_purchases = sd(purchases, na.rm = TRUE),
    median_purchases = median(purchases, na.rm = TRUE)
  )
print(purchases_summary)
```

A tibble: 2 × 5

	balance_tier	count	mean_purchases	sd_purchases	median_purchases
	<chr>	<int>	<dbl>	<dbl>	<dbl>
1	High Balance	682	1897.	4126.	409.
2	Low Balance	4800	846.	1261.	402.